Project Description

This project analyzes the distribution of alternative fuel vehicles across U.S. states to identify patterns in adoption, regional differences, and indicators of future growth. The analysis focuses on Electric Vehicles (EV), Plug-In Hybrid Electric Vehicles (PHEV), Hybrid Electric Vehicles (HEV), and other alternative fuels to understand where EV infrastructure investment would have the greatest impact. Using state-level vehicle counts and percentages, the project evaluates both current adoption and readiness indicators to support data-driven recommendations for policymakers and planners.

Summary of Analysis

The dataset was cleaned and aggregated using SQL to calculate total vehicle counts by fuel type and the percentage of each fuel type within every state. Visualizations were created in Tableau and included:

- A nationwide alternative fuel market share breakdown
- Gasoline, HEV, EV and PHEV % by state
- EV adoption by state
- A scatterplot comparing EV% vs. HEV%
- Identification of states with high hybrid adoption but low EV penetration

The analysis shows that gasoline remains the dominant fuel source in the United States. Alternative fuels are present in smaller but meaningful quantities, with HEV and EV showing the strongest growth. EV adoption is highest along the West Coast, while much of the Midwest and South remain low to moderate adopters. The scatterplot comparing EV percent and HEV percent highlights states where hybrid adoption is significantly higher than EV adoption, which signals strong consumer readiness for future electrification.

Key Insights

1. **EV adoption is regionally concentrated.** West Coast states show the highest EV percentages, which reflects early adoption behavior and established charging networks. The Midwest and South show lower EV penetration, which indicates early-stage markets.

2. **The U.S. alternative fuel market is dominated by Ethanol, HEV, and EV.** These fuels have the strongest nationwide presence due to availability, consumer familiarity, and existing support infrastructure. Other fuels such as propane, CNG, hydrogen, and methanol represent extremely small shares and are best grouped as Other Alternative Fuels.

3. **Gasoline remains dominant, but electrification is accelerating.** While gasoline vehicles make up much of the fleet, the growing presence of EVs and hybrids indicates a gradual shift toward cleaner technologies.

Recommendations

1. Prioritize EV infrastructure investment in Pennsylvania. Pennsylvania shows low EV adoption (~0.7%) but strong HEV presence (2.4%). With a large population and major interstate corridors, expanding charging infrastructure would accelerate EV adoption and support regional travel.

2. Invest in Minnesota as an emerging EV market. Minnesota has one of the highest hybrid adoption rates in the Midwest but relatively low EV penetration. This readiness gap suggests that infrastructure, not demand, is the primary barrier. Targeted investment would unlock significant future EV growth.

3. Expand infrastructure in North Carolina to support rapid population and EV growth. North Carolina combines moderate EV adoption, strong HEV presence, and one of the fastest-growing populations in the country. Infrastructure investment here would support both current demand and future expansion.